

David A. Plappert

Principal Cloud Architect · Consulting Architect · AWS Reliability Lead

Peoria, IL · (651) 894-4607 · david@davidplappert.com · [linkedin.com/in/davidplappert](https://www.linkedin.com/in/davidplappert) · github.com/davidplappert

US Patent 11,146,689, Cloud Architecture (2021) · 2× AWS City on a Cloud award (2016, 2018) · 19 years on AWS, since EC2 beta



Schedule 30 min →

SUMMARY

- Principal-level cloud architect with 19 years designing, building, and operating mission-critical AWS platforms; patent holder; two-time AWS innovation award winner
- Consulting-native: 10 years delivering parallel client engagements at a government health-tech consultancy — architecture I scoped and delivered grew one engagement by \$1M/yr and another by \$2M/yr; 18F (federal digital services, GSA) advisor
- Currently the sole Caterpillar-side AWS authority on **Vertex 3D**, Caterpillar's vendor-built real-time 3D platform powering aftermarket commerce — independent architecture review of an eight-figure vendor platform relationship
- Ships AI-augmented engineering in production today: Claude Code orchestrated CI/CD, agentic deployment workflows, MCP-based tool integration, multi-agent parallel orchestration
- Specializes in architecture reviews aligned to the AWS Well-Architected Framework, multi-account AWS architecture, high-availability and disaster-recovery design, observability, platform engineering, and migration strategy
- Industry depth:** Insurance, Payments, and Finance (10 yrs on state health insurance exchanges, \$500M/yr atomic payments, IRS / Stripe / EDI integrations); Manufacturing (Caterpillar aftermarket-commerce backbone); Healthcare and Federal Government (DCHBX, MA HCB, CoverME, DC DBH, 18F advisor); Retail and Point-of-Sale (Best Buy, billion-dollar BestBuy.com and POS systems on AWS); Media / Streaming (PBS video platform; RTMP / SRT ingest pipelines)

PROFESSIONAL EXPERIENCE

Caterpillar Inc. (via TEKsystems), Lead Cloud Engineer / AWS Reliability Lead

September 2025 – Present · Chillicothe, IL

- Sole Caterpillar-side AWS authority for **Vertex 3D**, Caterpillar's real-time 3D collaboration platform powering aftermarket commerce — ~\$3,000/minute in revenue when running, ~30 downstream publisher integrations; technical oversight of an **eight-figure vendor platform relationship**
- Independent architecture review authority: every monthly vendor-shipped AWS CDK release passes my architecture review and pre-deploy testing before production; typically remediate 2–3 defects per release before they can reach the platform
- Author of the technical findings behind Caterpillar's formal architecture-correctness escalations to the vendor; present findings and corrective requirements to Caterpillar business and engineering leadership
- Architected the platform's load-testing, monitoring, and alerting design; supported scaling from 2,000 to 8,000 concurrent users
- Designed and built a Claude Code agentic workflow that runs production deployments, pre-production dry-runs, automated bug detection, ticket filing, and end-to-end CI/CD pipeline orchestration — agents read architecture and open work, propose changes, file tickets, push code, and watch the deploy
- Parallel multi-agent orchestration against several stacks simultaneously, bridged into GitHub, AWS CLI, and the team's ticketing system through custom MCP servers and tool definitions I wrote

- Cross-team escalation authority across both AWS and Node.js stacks for Caterpillar's Publisher Distribution Team (PDT)
- Technical lead for Caterpillar's Node.js workstream on Vertex 3D; built and operate multiple production applications on the platform; mentoring junior Node.js engineer
- **Stack:** AWS (Fargate, ECS, S3, RDS / Aurora Serverless, Lambda, CloudWatch, IAM), AWS CDK, CloudFormation, Node.js, Java / JVM, GitHub Actions, Claude Code, Anthropic SDK, MCP

IdeaCrew (government health-tech consultancy), Principal Cloud Architect & DevOps Lead

2015 – 2025 · Remote

- Employee #3; led cloud architecture and DevOps across parallel client engagements worth >\$50M annually as the firm grew past 100 employees
- Client-facing lead architect for four state-government platforms — DC Health Benefit Exchange (DCHBX), Massachusetts Health Connector for Business, CoverME (Maine), and DC Department of Behavioral Health — owning architecture, security, roadmap, and delivery end to end
- Owned the AWS architecture that won **2x AWS City on a Cloud awards (2016, 2018)** on the DCHBX platform
- Grew engagements through delivery: scoped and solo-built a data warehouse spanning 8 databases (MongoDB BI Connector, MySQL, PostgreSQL) feeding Tableau — **+\$1M/yr contract expansion**; designed 300+ automated HIPAA- and SOC 2-compliant data jobs — **+\$2M/yr additional expansion**
- **18F (federal digital services, GSA) advisor** — AWS architecture consultation and workshops for multiple 18F chapters on state-government health exchange replication
- Migration strategy leadership: led multi-year state health insurance migrations from federal FedHub to state-based exchanges, worth tens of millions; team held the fastest FedHub transition record
- Multi-account AWS governance: managed \$1M/month cloud spend across 12 AWS and 3 Azure accounts with account-level isolation and cost optimization
- Designed distributed data consistency patterns across multi-region, high-availability state-exchange architecture — idempotent EDI processing, eventual-consistency reconciliation, and atomic payment workflows handling >\$500M annually under HIPAA, SOC 2, and IRS Publication 1075 compliance
- Led SOC 2 certification and annual audit processes, including disaster-recovery and business-continuity design; presented architecture decisions and audit outcomes to state-agency executives and client leadership
- Led monolith-to-microservices migration onto AWS EKS, ECS, and Lambda; final escalation point for production incidents across every client engagement, company-wide, backed by **New Relic** observability (APM, transaction tracing, alerting)
- **Stack:** Ruby on Rails, Node.js, Angular, AWS, Azure, EKS, ECS, Docker, Terraform, MongoDB, MySQL (Aurora Serverless), DynamoDB, PostgreSQL, Oracle

PBS, Cloud Engineer

2014

- Built and operated AWS-based RTMP video streaming platform with 99.9% uptime
- Implemented adaptive bandwidth modulation, materially increasing viewer watch time
- Managed a development team of two direct reports

Argonne National Laboratory, Senior Cloud Architect

2013

- Architected and delivered the lab's federal grant-application platform
- Streamlined grant workflows, increasing grant throughput and lab revenue

Best Buy, Senior Cloud DevOps

2012

- Owned cloud migration of **BestBuy.com and Point-of-Sale, billion-dollar revenue systems**, plus New Product Ingestion, from on-premises infrastructure to AWS; automated scaling and patching
- Built production services in **Ruby on Rails** and operated **Java / JVM** workloads across the BestBuy.com and Point-of-Sale stacks, with Cassandra in the BestBuy.com data layer
- Reduced infrastructure costs by over 50% while improving deployment speed
- Led a 6-member DevOps engineering team

Early career, web & infrastructure engineering

2007 – 2012

- Running production workloads on AWS since the EC2 beta; earlier web-platform and infrastructure engineering roles — detail available on request

EDUCATION & CREDENTIALS

- **US Patent 11,146,689**, Cloud Architecture (Issued October 12, 2021)
- **AWS City on a Cloud Award**, 2016 and 2018
- **AWS Certified Solutions Architect – Associate** (2019 – 2021)
- **18F advisor** (federal digital services / GSA), AWS architecture consultation during IdeaCrew tenure
- **19 years on AWS**, operating production workloads since EC2 beta
- **Education**: College coursework; self-taught engineer, production-trained on AWS since 2007

TECHNICAL SKILLS

- **Cloud & Infrastructure**: AWS (Fargate, ECS, EKS, S3, RDS, Lambda, CloudWatch, IAM, VPC, CDK, CloudFormation), multi-account architecture (AWS Organizations), Azure, GCP
- **Architecture & Reliability**: AWS Well-Architected Framework, distributed systems, event-driven architecture, microservices, multi-region, high availability, disaster recovery / business continuity, hybrid cloud, platform engineering, architecture and design reviews
- **Consulting & Client Leadership**: engagement scoping and expansion, pre-sales technical support, executive workshops and design reviews, multi-client parallel delivery, vendor architecture oversight, statements of work input, mentoring
- **AI Engineering & Agent Tooling**: Claude Code CLI for production engineering work, Anthropic API / SDK, multi-agent parallel orchestration, context engineering, MCP servers and custom tool definitions, AI-assisted CDK / infrastructure development, evaluation harnesses, RAG patterns
- **CI / CD & Developer Productivity**: GitHub Actions, Jenkins, AWS CodePipeline, Chef, agentic CI/CD orchestration, automated PR review, agent-assisted incident analysis and runbook automation
- **AI-SDLC Governance**: IP protection patterns, secret scanning, open-source license compliance, audit trail design (transferred from SOC 2 / HIPAA / IRS Pub 1075 background)
- **Infrastructure as Code & Containers**: AWS CDK, CloudFormation, Terraform, Docker, Kubernetes, EKS
- **Languages & Frameworks**: Ruby / Rails, Node.js / Express, JavaScript / TypeScript (React, Vue / Nuxt, Angular, Astro), Python, Bash, SQL, PHP (author); Java / JVM (operate on AWS); Serverless Framework
- **Databases**: PostgreSQL, MySQL (Aurora Serverless, TiDB, MariaDB), DynamoDB, Cassandra, MongoDB (MongoBI), Oracle, Redis
- **Video & Streaming**: RTMP / SRT live ingest, adaptive bitrate, low-latency pipelines on AWS
- **Observability**: New Relic (APM, tracing, alerting), CloudWatch, OpenTelemetry, distributed tracing
- **Compliance & Security**: HIPAA, SOC 2 certification and audit, IRS Publication 1075, security automation